A RULE SATISFYING THE BIVARIANT CONDITION FOR TRIVIALLY NORMAL BUNDLES FACTORS THROUGH RATIONAL EQUIVALENCE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The counted statement contains the author's own reference display environment at source line 6478 ('Very weak form of \cite[Theorem 17.1]{F}'), which the project typesets as a comment and whose definition comes from the preamble span over lines 8-12 of the shared preamble. The citation is inside the commented block and never executed.

Allowed standard prerequisites (editorial).

- The bivariant class definition and the Gysin operation on the level of cycles. Same chapter chow.tex, labels definition-bivariant-class (proof line 6490), definition-gysin-homomorphism (proof line 6498) and remark-gysin-on-cycles (proof line 6501). The first two fix the setting, and the remark is what lets the operation c cap - be evaluated on individual cycles, which the statement needs in order to make sense.
- The two fibre classes of a cycle on a product with the projective line. Same chapter chow.tex, label lemma-rational-equivalence-family, invoked at proof lines 6504 and 6515. Realises a rationally trivial cycle alpha as the difference of the two pullbacks of a single cycle beta on the product with the projective line, and later supplies the rational equivalence of the two pushes forward of c cap beta. That lemma is claimed separately as S14-algebra-rational-equiv-family.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original definition 0B76: sources/stacks-chow/pinned-original-source.tex, source lines 6087–6087. Navigation pointer for \ref{definition-bivariant-class} used inside the counted span (line 6490); the argument that consumes it is retained in the span.

Source reference. Original definition 02T8: sources/stacks-chow/pinned-original-source.tex, source lines 5069–5069. Navigation pointer for \ref{definition-gysin-homomorphism} used inside the counted span (line 6498); the argument that consumes it is retained in the span.

Source reference. Original lemma 02S4: sources/stacks-chow/pinned-original-source.tex, source lines 3335–3335. Navigation pointer for \ref{lemma-rational-equivalence-family} used inside the counted span (line 6504); the argument that consumes it is retained in the span.

Source reference. Original remark 0B6Z: sources/stacks-chow/pinned-original-source.tex, source lines 5237–5237. Navigation pointer for \ref{remark-gysin-on-cycles} used inside the counted span (line 6501); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 6481); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let $f : X \to Y$ be a morphism of schemes locally of finite type over S. Let $p \in \mathbf{Z}$. Suppose given a rule which assigns to every locally of finite type morphism $Y' \to Y$ and every k a map*

$$c \cap - : Z_k(Y') \longrightarrow \mathrm{CH}_{k-p}(X')$$

where $Y' = X' \times_X Y$, satisfying condition (3) of Definition 0B76 whenever $\mathcal{L}'|_{D'} \cong \mathcal{O}_{D'}$. Then $c \cap -$ factors through rational equivalence.

Proof. The statement makes sense because given a triple $(\mathcal{L}, s, i : D \to X)$ as in Definition 02T8 such that $\mathcal{L}|_D \cong \mathcal{O}_D$, then the operation i^* is defined on the level of cycles, see Remark 0B6Z. Let $\alpha \in Z_k(X')$ be a cycle which is rationally equivalent to zero. We have to show that $c \cap \alpha = 0$. By Lemma 02S4 there exists a cycle $\beta \in Z_{k+1}(X' \times \mathbf{P}^1)$ such that $\alpha = i_0^*\beta - i_\infty^*\beta$ where $i_0, i_\infty : X' \to X' \times \mathbf{P}^1$ are the closed immersions of X' over $0, \infty$. Since these are examples of effective Cartier divisors with trivial normal bundles, we see that $c \cap i_0^*\beta = j_0^*(c \cap \beta)$ and $c \cap i_\infty^*\beta = j_\infty^*(c \cap \beta)$ where $j_0, j_\infty : Y' \to Y' \times \mathbf{P}^1$ are closed immersions as before. Since $j_0^*(c \cap \beta) \sim_{rat} j_\infty^*(c \cap \beta)$ (follows from Lemma 02S4) we conclude. □